Non-Functional Requirements (HVPN-NFR-NNN)

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Revision: 1 **Last modified:** 2026-06-26T12:00:00Z

> **Document role.** This is the Volume 1 (Product & Requirements) nano-detail spec enumerating every HelixVPN **non-functional requirement (NFR)** — performance, scale, availability/HA, privacy, security, resource, and portability — each numbered `HVPN-NFR-NNN`. It is the quantified counterpart to the *seven non-negotiable principles* and the *Mullvad-parity matrix* of the Volume 1 overview (`00-product-scope-and-principles.md`). Where a value is a **target not yet proven on hardware** it is stated as a TARGET and marked `UNVERIFIED` per §11.4.6 (no-guessing) — a number that only a Phase-0/1 gate or a captured benchmark can confirm is never asserted as established fact here.
>
> **Document is a SPEC.** It states *what quality bars the product must meet and how each is verified*; it does not build the product. Every NFR carries a measurable target plus the §11.4.169 test type that proves it, so each row is falsifiable rather than aspirational (§11.4 / §11.4.169).
>
> **Evidence base.** Cross-references the Volume 1 overview (`00-product-scope-and-principles.md` — principles P1–P7, parity matrix F1–F17, decisions D1–D7, scope §10), the control-plane coordinator SLO (`v03-control-plane/svc-coordinator.md` §7 — convergence p99 < 1 s), the telemetry counts-only metric registry (`v03-control-plane/svc-telemetry.md` §3 — what may be measured), the no-logging invariant (`v05-security/no-logging-as-code.md` — the privacy NFRs as testable bars), and the transport ladder (`v02-data-plane/transport-selection-ladder.md`). Source research is cited by the same ids the overview uses (`[04_ARCH §N]`, `[04_P0]`, `[04_P1]`).

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1. How to read an NFR row

Every NFR is one row with five fixed fields, so each is independently testable (§11.4 — a quality bar with no measurement is a PASS-bluff at the requirements layer):

Field	Meaning
ID	`HVPN-NFR-NNN`, stable, append-only — the cross-doc reference key (§11.4.54 discipline applied to NFRs).
Statement	the qualitative requirement (one sentence).
Target	the measurable bar (number + unit + percentile/condition). `TARGET` = not yet proven on hardware (`UNVERIFIED` §11.4.6).
Verify by	the §11.4.169 test type whose captured evidence proves it (unit / integration / e2e / performance / soak-stress / chaos / security / memory / benchmarking).
Priority	`P0` (Phase-0 gate / make-or-break), `MVP` (Phase-1 DoD), `P2`/`P3` (later phase), `ALWAYS` (every phase).

Priority bands map to the §10 scope spine of the overview: a `P0` NFR is a make-or-break Phase-0 exit gate; an `MVP` NFR is part of the 8-criteria Definition-of-Done; a `P2`/`P3` NFR binds when its phase opens.

> **Honest-number rule (§11.4.6).** Several targets below originate from the refined research as *design targets* (e.g. the iOS NE memory ceiling, the ⩾80%-bare-link throughput gate). They are reproduced as TARGETs and marked `UNVERIFIED` — the Phase-0 gate or the named benchmark is what converts a TARGET into an established fact. No target is restated here as if already measured.

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2. Performance NFRs (HVPN-NFR-001…)

Performance is dominated by two budgets: **datapath throughput/latency** (the Rust data plane, doc 01) and **control-plane convergence** (the coordinator, `svc-coordinator.md`).

ID	Statement	Target	Ver
HVPN-NFR-001	Plain-UDP WireGuard datapath sustains near-bare-link throughput client→gateway→connector-LAN.	⩾ **80 %** of bare-link throughput (Phase-0 gate G1) — TARGET `UNVERIFIED`	per ben (ipe the
HVPN-NFR-002	MASQUE/QUIC obfuscated datapath retains usable throughput through a DPI UDP block.	⩾ **50 %** of plain-UDP throughput (Phase-0 gate G2) — TARGET `UNVERIFIED`	per ben
HVPN-NFR-003	Control-plane convergence: a policy/ topology change reaches every affected agent's `WatchNetworkMap` stream.	**p99 < 1 s** event→delta-on-wire (`helix_reconcile_seconds`)	per (syn poli load
HVPN-NFR-004	Device revoke is enforced (peer-removal delta on wire + stream force-close).	**< 1 s** revoke-event→`sub.cancel()` (`helix_revoke_enforce_seconds` p99)	per + se
HVPN-NFR-005	WireGuard handshake completes within a bounded latency on a healthy link.	handshake-complete **TARGET ⩽ 1 s** typical; rung success = WG handshake, not carrier connect — `UNVERIFIED` exact bound	inte (har tim cap
HVPN-NFR-006	Transport-ladder escalation to a working rung is bounded per attempt (no unbounded hang).	per-rung budget enforced; cursor advances monotonically (L-I8); total ⩽ Σ rung budgets — `UNVERIFIED` calibrated value	inte (lad tran trac
HVPN-NFR-007	Datapath added latency (encryption + obfuscation overhead) stays within an obfuscation-aware budget.	plain-UDP overhead **TARGET** near-zero; MASQUE adds QUIC/ H3 framing cost, bounded — `UNVERIFIED` measured value	ben (pe late hist
HVPN-NFR-008	MTU is chosen so WG-in-transport never fragments on a 1500-byte path.	WG MTU **1420** plain; transport-specific reductions for MASQUE/ Shadowsocks documented per rung	inte (pat pro frag asso
HVPN-NFR-009	Admin-perceived "policy edit reflected" end-to-end (REST→compile→activate→bus→delta).	**< 1 s** as the MVP DoD-5 criterion (superset of NFR-003, per-hop timers)	e2e aut

> **Note on NFR-003 vs NFR-009 (§11.4.6).** NFR-003 measures the coordinator's `event-receive → stream.Send` segment only; NFR-009 is the wider admin-edit path the DoD asserts and is a *superset* — they are tracked as distinct series (`svc-coordinator.md` §7.1, `svc-telemetry.md` §6.1), not conflated.

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3. Scale NFRs (HVPN-NFR-100…)

Scale targets are stated as MVP single-node bars plus Phase-2 fleet bars. Where the MVP number is a sizing target not yet load-proven it is `UNVERIFIED`.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-100	A single self-hosted gateway pod serves a homelab-scale fleet of concurrent agents.	**TARGET ⩾ 10 k** concurrent `WatchNetworkMap` streams per coordinator (the soak-test figure) — `UNVERIFIED`	soak-stress (24 h, N=10 k streams)	MVP
HVPN-NFR-101	Coordinator memory stays bounded under sustained stream load (no leak).	`process_resident_memory_bytes` slope ≈ **0** over 24 h @ 10 k streams	soak-stress / memory	MVP
HVPN-NFR-102	The minimal-affected-set fan-out keeps per-event work small regardless of tenant size.	`helix_fanout_affected_nodes` ≪ **tenant size** for a small change	unit + performance	MVP
HVPN-NFR-103	One gateway supports many joined private networks	**TARGET** connectors/tenant bounded only by overlay-IP space (ULA /48, D4) — `UNVERIFIED` concrete ceiling	integration (multi-connector overlay)	MVP

ID	Statement	Target	Verify by	Priority
	(connectors) per tenant.			
HVPN-NFR-104	One user reaches N joined networks (the multi-network differentiator X1).	`1 user → N nets`, N bounded by policy + IPAM, not architecture	e2e (multi-network reach)	MVP
HVPN-NFR-105	Reconnect storm (mass gateway flap) is absorbed without deadlock or unbounded buffering.	10 k simultaneous reconnects: bounded send queues, no deadlock, drops counted not buffered	stress (reconnect-storm)	MVP
HVPN-NFR-106	Per-tenant isolation does not let one tenant's churn block another.	per-tenant `tenantState` sharding; one tenant's lock never held across another	concurrency / race-deadlock	MVP
HVPN-NFR-107	Control plane scales horizontally to stateless multi-replica coordinators.	coordinators stateless (graph rebuilt from Postgres+events on boot); N replicas, no sticky state	integration (multi-replica)	P2
HVPN-NFR-108	Event backbone scales from single-node Redis Streams to multi-region NATS JetStream.	bus interface (D3) transport-agnostic; swap is mechanical, not a redesign	integration (bus swap)	P2

4. Availability & HA NFRs (HVPN-NFR-200…)

The load-bearing availability property is **fail-static** (P1): a control-plane outage is never a connectivity outage.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-200	Existing tunnels keep forwarding while the control plane is down (fail-static, P1).	**0** tunnel drops on control-plane outage; edge forwards from cached verdict map	chaos (kill control plane mid-traffic)	ALWAYS
HVPN-NFR-201	A Postgres/ Redis blip does not restart the gateway process.	liveness probe dependency-free; readiness gates new traffic only, never kills serving streams	chaos (DB/ Redis blip)	MVP
HVPN-NFR-202	Presence backend (Redis) loss degrades gracefully, never errors a tunnel (fail-static).	`IsOnline→ErrPresenceUnavailable`; coordinator assumes reachable-via-relay degraded; no tunnel error	chaos (drop Redis mid-stream)	MVP
HVPN-NFR-203	A crashed coordinator consumer loses no events (no-work-loss).	XAutoClaim reclaim of stalled PEL entries; **0** lost events across a kill	chaos (kill consumer mid-apply)	MVP
HVPN-NFR-204	A coordinator restart is transparent to connected agents (resume by known_version).	agents resume from `known_version` against the rebuilt-from-Postgres graph; snapshot fallback on ring gap	chaos / integration (restart mid-stream)	MVP
HVPN-NFR-205	Multi-region HA: stateless coordinators + Patroni Postgres + NATS JetStream survive a region loss.	**TARGET** region-failover within an RTO/RPO budget — `UNVERIFIED` (Phase-2 DR runbook owns the number)	chaos (region-failover drill)	P2

ID	Statement	Target	Verify by	Priority
HVPN-NFR-206	Gateway failover re-points agents to a healthy gateway endpoint.	`gateway.failover` event → every tenant agent gets an updated endpoint within the convergence SLO	integration / chaos	P2
HVPN-NFR-207	Poison events never spin the consumer; they route to a DLQ after a delivery-count ceiling.	delivery-count ⩾ cap → DLQ + alert; **no** infinite retry	chaos / integration (DLQ)	MVP

> **Availability SLO numbers are deferred to the DR doc (§11.4.6).** The concrete uptime %, RTO, and RPO budgets live in `v06-deploy/disaster-recovery.md` (which closes ledger gap G1); reproducing a specific "99.9 %" here would be a fabricated number — NFR-205 names the budget's *owner*, not a guessed value.

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5. Privacy NFRs (HVPN-NFR-300…)

Privacy is the product's reason to exist; these NFRs make the no-logging guarantee **testable** rather than promissory. They are bound by `no-logging-as-code.md` — what may be measured is constrained by what may exist.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-300	No durable connection/traffic/packet/flow/session table exists anywhere in the schema (P7, S6).	schema-lint **exit 0** against migrations AND the live `information_schema`	security (schema-lint, pre-build + post-deploy runtime signature)	ALWAYS
HVPN-NFR-301	The schema-lint is not a tautology (it actually catches a planted flow table).	planted `flows(src,dst,bytes,ts)` migration → lint **FAILs**; removed → PASSes (paired §1.1)	meta-test (§1.1 golden-bad/good)	ALWAYS
				MVP

ID	Statement	Target	Verify by	Priority
HVPN-NFR-302	The event bus carries no traffic shape (C3 on the bus).	zero src/dst/bytes/dns fields in any event payload struct; payload-lint green	security (payload-lint, build + publish-time)	
HVPN-NFR-303	Audit records control actions only, never traffic; the `meta` jsonb cannot smuggle a flow.	closed action vocabulary; `meta` key matched against forbidden regex → `ErrAuditMetaShape`	unit + integration (audit meta guard)	MVP
HVPN-NFR-304	Live presence is ephemeral (Redis TTL), never persisted to Postgres.	`presence:{tenant}:{device}` TTL = 3× heartbeat; **no** durable session row; loss of Redis loses no durable state	integration (presence TTL/expiry)	MVP
HVPN-NFR-305	The one durable presence derivative (`last_seen_at`) cannot reconstruct a session timeline.	coarsened to **⩾ 5 min** (`LAST_SEEN_COARSEN`), carries no destination	unit + integration	MVP
HVPN-NFR-306	`/metrics` exposes aggregate counters only — no per-user, per-flow, per-destination, or per-tenant-label series.	every collector's label set ⊆ allow-list; **no** `tenant_id`/`device_id`/`*_ip` label	unit (label-cardinality audit)	MVP
HVPN-NFR-307	Anonymous identity is supported (no email, no SSO required).	a tenant mints device enroll tokens with `users.email = NULL`, `oidc_sub = NULL`; no reverse-link to a human	integration (anonymous enroll)	MVP
HVPN-NFR-308	The no-logging		e2e Challenge	MVP

ID	Statement	Target	Verify by	Priority
	guarantee holds as a runtime signature, not a source grep (§11.4.108).	schema-lint green against the **deployed** DB, asserted post-deploy	(deployed-DB lint)	

> **Honest boundary (§11.4.6).** These NFRs guarantee *durable-store* + *bus* + *audit* absence of traffic data. They do **not** claim a hot-compromised running gateway holds nothing in RAM (live presence keys + aggregate counters are the residual surface T5 in `no-logging-as-code.md` §14). The privacy NFR set is "no durable user?destination correlation ever exists," never "an attacker with root on a live box sees nothing."

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6. Security NFRs (HVPN-NFR-400…)

ID	Statement	Target	Verify by	Priority
HVPN-NFR-400	WireGuard's audited crypto core is never forked; obfuscation is a layer *under* it (P2).	WG Noise IK / Curve25519 / ChaCha20-Poly1305 untouched; CI lint forbids edits to WG crypto primitives	security + meta-test	ALWAYS
HVPN-NFR-401	Edges are outbound-only; no private network needs an inbound port (P3).	only the gateway listens publicly; connectors/clients dial out	security (no-inbound assertion)	ALWAYS
HVPN-NFR-402	Default-deny, need-to-know: a node only ever learns peers a compiled rule grants (C4).	a node with no `VisibleTo` entry sees **zero** peers; deny-unauthorized peer never appears in any snapshot	e2e + meta-test (skip-filter mutation FAILs)	MVP
HVPN-NFR-403	Device identity is mTLS with short-lived	short-lived device cert; revoke enforced **< 1 s**	security	MVP

ID	Statement	Target	Verify by	Priority
	certs and instant revocation.	(NFR-004); impersonation (`device_id`≠cert) rejected		
HVPN-NFR-404	Kill-switch prevents any leak if the tunnel drops.	**0** packets off-tunnel on tunnel drop (OS firewall state machine)	security / e2e (leak test on drop)	MVP
HVPN-NFR-405	DNS-leak protection forces tunnel DNS and blocks plaintext off-tunnel :53.	**0** plaintext DNS off-tunnel	security / e2e (DNS-leak test)	MVP
HVPN-NFR-406	The device private WG key never leaves the device.	key generated on-device at enrollment; only the public key is registered (C6)	integration + security	MVP
HVPN-NFR-407	Post-quantum handshake is hybrid (PQ pre-shared layer), never PQ-only.	ML-KEM/FIPS-203 PSK layered on WG; classical handshake retained — hybrid-never-PQ-only	security + integration	P2
HVPN-NFR-408	Multi-tenant isolation is enforced at the database (RLS), not only in app code.	`FORCE ROW LEVEL SECURITY`; tenant A cannot read tenant B even with a crafted query as `helix_app`	security / integration (RLS)	MVP
HVPN-NFR-409	Every change crosses the mandatory anti-bluff test gauntlet before release.	each §11.4.169 type present + paired §1.1 mutation that flips its gate RED	meta-test (per-gate mutation)	ALWAYS
HVPN-NFR-410	Censorship-evasion: the ladder reaches a working obfuscated	escalation reaches MASQUE (or a later rung) and completes a WG	e2e (DPI-sim escalation)	MVP

ID	Statement	Target	Verify by	Priority
	transport under DPI/ UDP-block.	handshake under a DPI sim		

7. Resource NFRs (HVPN-NFR-500…)

The single hardest constraint in the whole product is the **iOS Network Extension memory ceiling —** it shapes decision D2 (Rust core).

ID	Statement	Target	Verify by	Pr
HVPN-NFR-500	The Rust client core runs inside the iOS `NEPacketTunnelProvider` memory budget with headroom (make-or-break).	working-set under the iOS NE ceiling with **⩾ 30 % headroom** (Phase-0 gate G3) — the historical ~15 MB ceiling is a measured constraint, **TARGET `UNVERIFIED` until G3 on-device**	memory (on-device NE soak)	P0
HVPN-NFR-501	Mobile client is battery-sensitive: push-don't-poll, no busy loops.	event-driven reconcile (no cron/ poll); **TARGET** measurable battery delta vs idle — `UNVERIFIED`	benchmarking (battery/CPU on-device)	M
HVPN-NFR-502	The client core fits a small binary footprint on size-constrained platforms.	**TARGET** core `.so`/`.a` size budget per platform — `UNVERIFIED` concrete bytes	benchmarking (artifact-size check)	M
HVPN-NFR-503	Project procedures (build/test/tooling) never exceed 60 % of host RAM (§12.6).	host-side resource ceiling **⩽ 60 %** total RAM	benchmarking (host resource sampler §11.4.24)	A
HVPN-NFR-504	The control-plane process holds bounded memory under sustained load (no leak).	`process_resident_memory_bytes` slope ≈ 0 over 24 h (= NFR-101, server side)	soak-stress / memory	M
HVPN-NFR-505	Telemetry samplers (observers) stay lightweight (Heisenberg constraint).	a resource sampler stays under its own RSS/CPU budget (§11.4.24: < 50 MB RSS, < 5 % CPU)	benchmarking	M
HVPN-NFR-506	Slow-consumer streams are dropped, not buffered unboundedly	per-stream `SEND_QUEUE_CAP` bound; overflow → drop + reconnect-with-snapshot	stress / memory	M

ID	Statement	Target	Verify by	Pr
	(back-pressure not growth).			

> **NFR-500 is the program's pivot (§11.4.6).** The overview names G3 as the literal decider for D2: if Rust cannot fit the iOS NE ceiling with headroom, D2 *and the whole client strategy* re-open. The ~15 MB figure is reproduced as the historical measured constraint and marked `UNVERIFIED` until the G3 on-device capture exists — it is never asserted here as a settled fact.

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8. Portability NFRs (HVPN-NFR-600…)

HelixVPN targets **8 platforms on one shared codebase** (Rust core + Flutter UI

- per-platform tunnel shims). Portability NFRs are per-platform reachability bars, phased per §10.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-600	All three apps build from one Flutter tree via the `runHelixApp(flavor,…)` entrypoint.	flavors `{Access, Connector, Console}`; Console omits `core_ffi` (no tunnel core)	integration (per-flavor build)	MVP
HVPN-NFR-601	The Rust core is shared byte-for-byte across Client, Connector, and Gateway edge (P5).	one `helix-transport` crate; wrap-on-client == unwrap-on-edge	integration (shared-core parity)	MVP
HVPN-NFR-602	MVP Access app runs on iOS, Android, and Linux.	three platforms green on the 8-criteria DoD	e2e / full-automation	MVP
HVPN-NFR-603	Connector daemon runs headless on Linux/ Windows/macOS (+	headless daemon + optional slim	integration	MVP

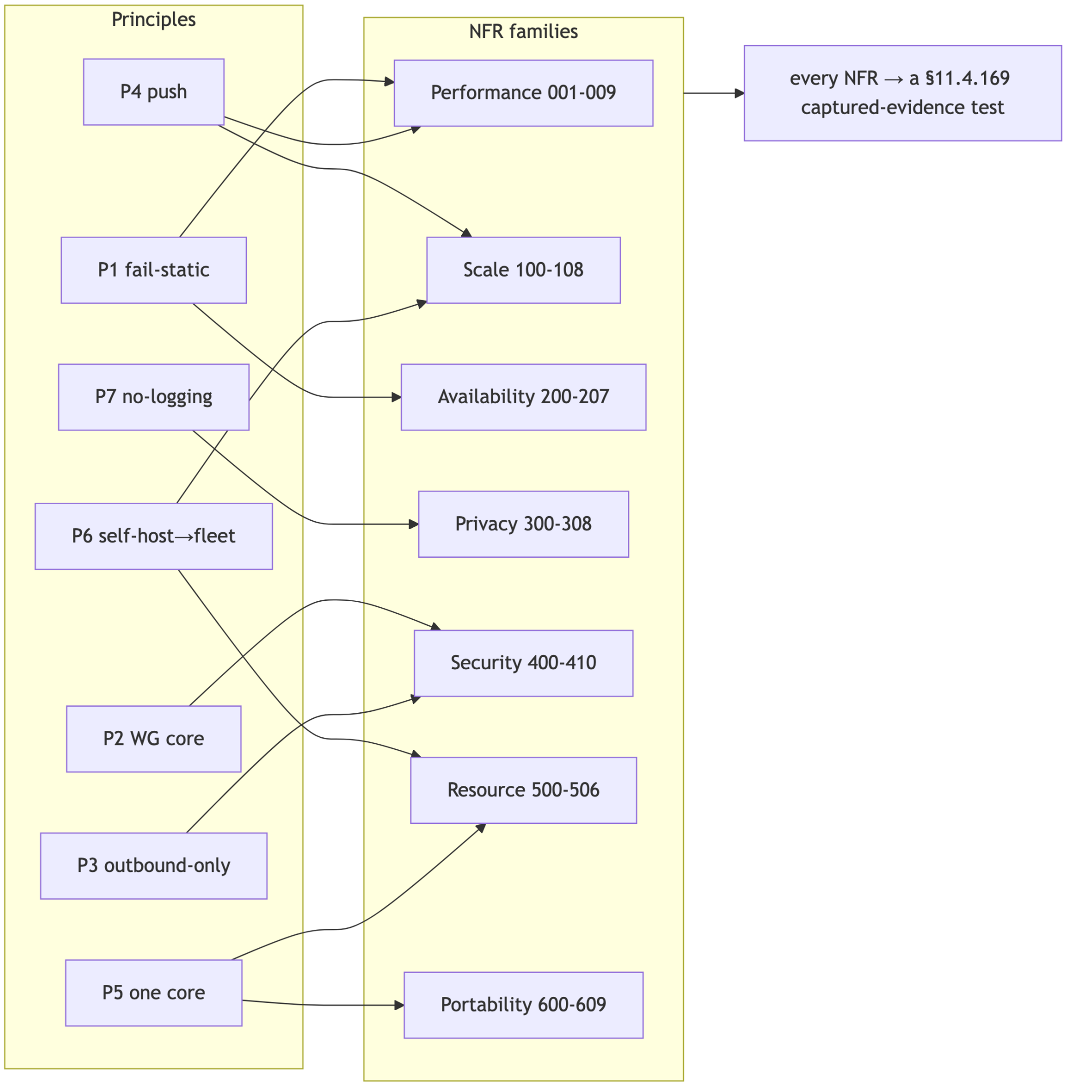
ID	Statement	Target	Verify by	Priority
	Android/embedded for appliances).	UI; advertise/ route mode		
HVPN-NFR-604	Console runs responsively as Web and as a desktop Flutter build.	one Flutter build serves Web + Desktop; admin flavor, API-client only	e2e (responsive Web + desktop)	MVP
HVPN-NFR-605	Phase-2 desktop apps reach Windows (`wireguard–nt`+service) and macOS (NE).	Windows + macOS Access apps green	e2e	P2
HVPN-NFR-606	Phase-3 extends to HarmonyOS NEXT and Aurora OS (the biggest platform risk).	native tunnel-shim work green on both — **TARGET `UNVERIFIED`** (no incumbent precedent)	e2e (per-platform shim)	P3
HVPN-NFR-607	The Web build is honestly scoped: Console + an *optional* browser-scoped WASM MASQUE proxy, never a system VPN.	no TUN device in-browser; the WASM proxy proxies the browser's own traffic only	e2e + documentation honesty check	P3
HVPN-NFR-608	Every UI surface ships light + dark variants and passes visual-regression (§11.4.162 OpenDesign).	light+dark per component; no overlap/ overlay; visual-regression green	UI / visual-regression	P3
HVPN-NFR-609	Cross-platform parity: any platform-specific primitive has a per-OS equivalent or an honest gap (§11.4.81).	`uname –s`-dispatched equivalents or documented kernel-gap SKIP — never a silent	per-OS test branches	ALWAYS

ID	Statement	Target	Verify by	Priority
		Linux-only pass		

9. NFR → principle → verification traceability

Each NFR family traces to the overview's non-negotiable principles (P1–P7) and the parity matrix (F1–F17 / X1–X5), so no quality bar floats unanchored (§11.4.25 coverage discipline).

NFR family	Principle(s)	Parity / differentiator rows	Primary §11.4.169 test types
Performance (001–009)	P1 (fail-static keeps Go off hot path), P4 (push)	F1, F2, F7	performance, benchmarking, integration
Scale (100–108)	P4, P6 (self-host→fleet)	X3	soak-stress, stress, concurrency
Availability/ HA (200–207)	P1 (fail-static)	X3	chaos, integration
Privacy (300–308)	P7 (no-logging by construction)	F14, F15	security (schema/ payload lint), meta-test
Security (400–410)	P2 (WG core), P3 (outbound-only)	F11, F12, F13, F16, F17, X2, X4	security, e2e, meta-test
Resource (500–506)	P5 (one core), P6	F1, X5	memory, benchmarking
Portability (600–609)	P5 (one core, reused)	X5	integration, e2e, UI/visual-regression



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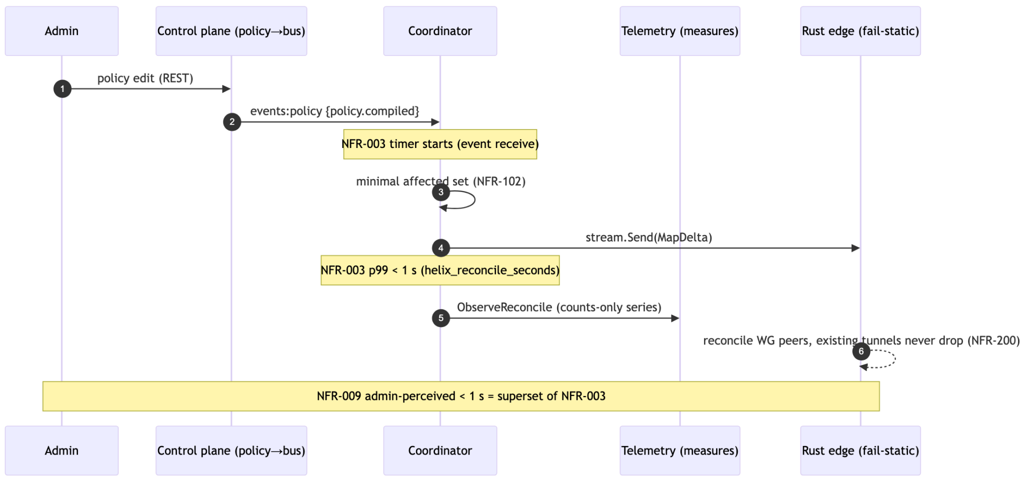
10. The convergence-NFR dependency chain

NFR-003 (p99 < 1 s convergence) and NFR-004 (< 1 s revoke) are the two headline real-time bars; they depend on a chain of lower NFRs holding, and the telemetry service makes each link a falsifiable series (counts/timing only, never traffic — `svc-telemetry.md` §3).



The dependency order (each lower NFR is a precondition of the SLO above it):

1. **NFR-101** / **NFR-506** (bounded memory + back-pressure) — without these the coordinator could not hold 10 k streams to *deliver* a sub-second delta.
2. **NFR-102** (minimal-affected-set) — a tenant-wide broadcast on every change would blow the 1 s budget; the minimal set keeps per-event work small.
3. **NFR-203** / **NFR-207** (no-work-loss + DLQ) — a stuck/lost event never converges at all, so reliability is a precondition of latency.
4. **NFR-200** (fail-static) — convergence is a *coordination* property; even when it is breached, connectivity (the edge datapath) must not be.

> **§11.4.6 closing note.** Every numeric target in §2–§8 marked `UNVERIFIED` is owned by a specific Phase-0/1/2 gate or named benchmark (G1, G2, G3, G4, the soak/stress suite, the DR runbook). The verification column is the contract: an NFR is "met" only when its named test produces captured evidence (§11.4.5/§11.4.69), never when the number is merely written down here.

Sources verified

- `docs/research/mvp/final/00-product-scope-and-principles.md` — principles P1–P7 (§8), Mullvad-parity matrix F1–F17 + X1–X5 (§9), phase scope + exit gates G1–G6 + 8-criteria MVP DoD (§10), decisions D1–D7 incl. the iOS NE ceiling shaping D2 (§11), the "fully-responsive-web ≠ system VPN" honesty marker (§2.2).
- `docs/research/mvp/final/v03-control-plane/svc-coordinator.md` §7 — the convergence < 1 s SLO, `helix_reconcile_seconds`, revoke < 1 s, the minimal-affected-set fan-out, the 24 h no-leak soak.

- `docs/research/mvp/final/v03-control-plane/svc-telemetry.md` §3 — the counts-only Prometheus registry, the forbidden-label cardinality bound (no `tenant_id`/`device_id`/ip labels), the SLO→alert mapping; §10 the test matrix.
- `docs/research/mvp/final/v05-security/no-logging-as-code.md` — the no-logging invariant as testable NFRs (schema-lint, payload-lint, audit meta guard, ephemeral presence, coarse `last_seen_at`, the §11.4.108 runtime signature), the honest-boundary residual risk T5.
- `docs/research/mvp/final/v02-data-plane/transport-selection-ladder.md` — the escalation-ladder budgets, monotonic-cursor invariant (L-I8), and aggregate no-per-user telemetry (I5) behind NFR-006 / NFR-410.

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — every unproven target marked `UNVERIFIED` and owned by a named gate/benchmark, never asserted as measured fact), §11.4.169 (each NFR names the test type whose captured evidence proves it), §11.4.25 (coverage ledger discipline — §9 traceability), §11.4.54 (stable append-only `HVPN-NFR-NNN` ids), §11.4.81 (cross-platform parity — NFR-609), §11.4.108 (runtime signature — NFR-308), §11.4.65/§11.4.153 (HTML+PDF[+DOCX] exports follow in refinement).